

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): LW502, LWU502

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.04.0000-b007	2.01.0001-b001	1.2.2

Version History

Module Version	Date	Notes
1_0_3_0	02/15/2017	Fixed connection diagram.
1_0_1_0	12/29/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:
InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5

Control Commands, States, and Qualifiers:

AspectRatio	'Normal' '16:10' 'Zoom'	'4:3' '14:9'	'16:9' 'Native'
AudioMute	'On'	'Off'	
AutoImage	None		
AVMute	'On'	'Off'	
ClosedCaptionsChannel	'1' – '4'		
ClosedCaptionsDisplay	'Off'	'On'	'Auto'
ClosedCaptionsMode	'Captions'	'Text'	
EcoMode	'Normal'	'Eco'	
Freeze	'On'	'Off'	
Input	'LAN' 'HDBaseT'	'HDMI 1' 'Video'	'HDMI 2' 'Computer In'
PbyPMainArea	'Left'	'Right'	
PbyPMainSize	'Small'	'Middle'	'Large'
PbyPPIPFrameLock	'Left / Primary'	'Right / Secondary'	
PbyPPIPModeSelect	'Off'	'PbyP'	'PIP'
PbyPSource	'Computer In' 'HDBaseT'	'HDMI 1' 'Video'	'HDMI 2'
['Area']	'Left'	'Right'	
PbyPSwap			
PictureMode	'Standard' 'Dynamic' 'Whiteboard' 'User-1'	'Natural' 'Board(Black)' 'Daytime' 'User-2'	'Cinema' 'Board(Green)' 'Dicom Sim' 'User-3'
PIPMMainArea	'Primary'	'Secondary'	
PIPPosition	'Top Left' 'Bottom Right'	'Top Right'	'Bottom Left'
PIPSource	'Computer In' 'HDBaseT'	'HDMI 1' 'Video'	'HDMI 2'
['Area']	'Primary'	'Secondary'	
Power	'On'	'Off'	
VideoMute	'On'	'Off'	
Volume	'Up'	'Down'	

['Input Type']	'Computer In' 'HDMI 2'	'LAN' 'HDBaseT'	'HDMI 1' 'Video'
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Status Available:

AspectRatio	'Normal' '16:10' 'Zoom'	'4:3' '14:9'	'16:9' 'Native'
AudioMute	'On'	'Off'	
AVMute	'On'	'Off'	
ClosedCaptionsChannel	'1' – '4'		
ClosedCaptionsDisplay	'Off'	'On'	'Auto'
ClosedCaptionsMode	'Captions'	'Text'	
ConnectionStatus	'Connected'	'Disconnected'	
DeviceStatus	'Normal' 'Lamp Error' 'Cold Error' 'AC Blackout Error'	'Cover Error' 'Temp Error' 'Filter Error'	'Fan Error' 'Air Flow Error' 'Shutter Error'
EcoMode	'Normal'	'Eco'	
FilterUsage	Hours		
Freeze	'On'	'Off'	
Input	'LAN' 'HDBaseT'	'HDMI 1' 'Video'	'HDMI 2' 'Computer In'
LampUsage	Hours		
PbyPMainArea	'Left'	'Right'	
PbyPMainSize	'Small'	'Middle'	'Large'
PbyPPIPFrameLock	'Left / Primary'	'Right / Secondary'	
PbyPPIPModeSelect	'Off'	'PbyP'	'PIP'
PbyPSource	'Computer In' 'HDBaseT'	'HDMI 1' 'Video'	'HDMI 2'
['Area']	'Left'	'Right'	
PictureMode	'Standard' 'Dynamic' 'Whiteboard' 'User-1'	'Natural' 'Board(Black)' 'Daytime' 'User-2'	'Cinema' 'Board(Green)' 'Dicom Sim' 'User-3'
PIPMMainArea	'Primary'	'Secondary'	
PIPPosition	'Top Left' 'Bottom Right'	'Top Right'	'Bottom Left'
PIPSource	'Computer In'	'HDMI 1'	'HDMI 2'

Global Scriptor Module Communication Sheet

	'HDBaseT'	'Video'	
['Area']	'Primary'	'Secondary'	
Power	'On'	'Off'	'Cooling Down'
VideoMute	'On'	'Off'	
VolumeStatus	0 –		
['Input Type']	'Computer In'	'LAN'	'HDMI 1'
	'HDMI 2'	'HDBaseT'	'Video'

Cable and Adapter Requirements:

Captive Screw to Female DB9 RS-232 Serial Cable

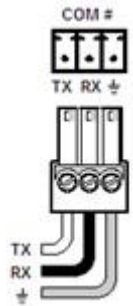
Notes for the Device:

To ensure that the projector can be remotely powered on, the Standby Mode option must be set to “Normal”.
This setting can be found in the menu under SETUP > STANDBY MODE

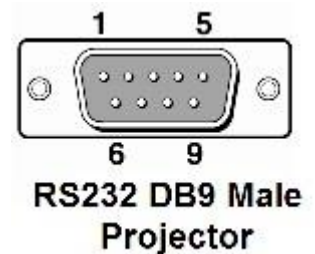
Serial communication:

Port Type: RS-232
Baud Rate: 19200
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram

Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND

**General Notes:**

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	Yes
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

Aspect Ratio 14:9	\xBE\xEF\x03\x06\x00\xCE\xD6\x01\x00\x08 \x09\x00
Aspect Ratio 16:10	\xBE\xEF\x03\x06\x00>\xD6\x01\x00\x08 \x0A\x00
Aspect Ratio 16:9	\xBE\xEF\x03\x06\x00\x0E\xD1\x01\x00\x08 \x01\x00
Aspect Ratio 4:3	\xBE\xEF\x03\x06\x00\x9E\xD0\x01\x00\x08 \x00\x00
Aspect Ratio Native	\xBE\xEF\x03\x06\x00^\xD7\x01\x00\x08 \x08\x00
Aspect Ratio Normal	\xBE\xEF\x03\x06\x00^\xDD\x01\x00\x08 \x10\x00
Aspect Ratio Zoom	\xBE\xEF\x03\x06\x00\x9E\xC4\x01\x00\x08 0\x00
Audio Mute Off	\xBE\xEF\x03\x06\x00F\xD3\x01\x00\x02 \x00\x00
Audio Mute On	\xBE\xEF\x03\x06\x00\xD6\xD2\x01\x00\x02 \x01\x00
Auto Image None	\xBE\xEF\x03\x06\x00\x91\xD0\x06\x00\x0A \x00\x00
AV Mute Off	\xBE\xEF\x03\x06\x00\xFE\xF0\x01\x00\xA0 \x00\x00
AV Mute On	\xBE\xEF\x03\x06\x00n\xF1\x01\x00\xA0 \x01\x00
Closed Captions Channel 1	\xBE\xEF\x03\x06\x00\xD2b\x01\x00\x027\x01\x00
Closed Captions Channel 2	\xBE\xEF\x03\x06\x00"b\x01\x00\x027\x02\x00
Closed Captions Channel 3	\xBE\xEF\x03\x06\x00\xB2c\x01\x00\x027\x03\x00
Closed Captions Channel 4	\xBE\xEF\x03\x06\x00\x82a\x01\x00\x027\x04\x00
Closed Captions Display Auto	\xBE\xEF\x03\x06\x00\x9Ac\x01\x00\x007\x02\x00
Closed Captions Display Off	\xBE\xEF\x03\x06\x00\xFAB\x01\x00\x007\x00\x00
Closed Captions Display On	\xBE\xEF\x03\x06\x00jc\x01\x00\x007\x01\x00
Closed Captions Mode Captions	\xBE\xEF\x03\x06\x00\x06c\x01\x00\x017\x00\x00
Closed Captions Mode Text	\xBE\xEF\x03\x06\x00\x96b\x01\x00\x017\x01\x00
Eco Mode Eco	\xBE\xEF\x03\x06\x00\xAB"\x01\x00\x003\x01\x00
Eco Mode Normal	\xBE\xEF\x03\x06\x00;#\x01\x00\x003\x00\x00
Freeze Off	\xBE\xEF\x03\x06\x00\x83\xD2\x01\x00\x020\x00\x00
Freeze On	\xBE\xEF\x03\x06\x00\x13\xD3\x01\x00\x020\x01\x00
Input Computer In	\xBE\xEF\x03\x06\x00\xFE\xD2\x01\x00\x00 \x00\x00
Input HDBaseT	\xBE\xEF\x03\x06\x00\xAE\xDE\x01\x00\x00 \x11\x00
Input HDMI 1	\xBE\xEF\x03\x06\x00\x0E\xD2\x01\x00\x00 \x03\x00
Input HDMI 2	\xBE\xEF\x03\x06\x00n\xD6\x01\x00\x00 \x0D\x00
Input LAN	\xBE\xEF\x03\x06\x00\xCE\xD5\x01\x00\x00 \x0B\x00
Input Video	\xBE\xEF\x03\x06\x00n\xD3\x01\x00\x00 \x01\x00
PbyP Main Area Left	\xBE\xEF\x03\x06\x00z&\x01\x00\x13#\x00\x00
PbyP Main Area Right	\xBE\xEF\x03\x06\x00\xEA'\x01\x00\x13#\x01\x00
PbyP Main Size Large	\xBE\xEF\x03\x06\x00\x92G\x01\x00\x11#\x81\x00
PbyP Main Size Middle	\xBE\xEF\x03\x06\x00\x02F\x01\x00\x11#\x80\x00
PbyP Main Size Small	\xBE\xEF\x03\x06\x00\xF2\x07\x01\x00\x11#\x7F\x00
PbyP PIP Frame Lock Left / Primary	\xBE\xEF\x03\x06\x00J'\x01\x00\x17#\x00\x00
PbyP PIP Frame Lock Right / Secondary	\xBE\xEF\x03\x06\x00\xDA&\x01\x00\x17#\x01\x00
PbyP PIP Mode Select Off	\xBE\xEF\x03\x06\x00>&\x01\x00\x10#\x00\x00
PbyP PIP Mode Select PbyP	\xBE\xEF\x03\x06\x00\xAE'\x01\x00\x10#\x01\x00
PbyP PIP Mode Select PIP	\xBE\xEF\x03\x06\x00^\x01\x00\x10#\x02\x00
PbyP Source Computer In Area Left	\xBE\xEF\x03\x06\x00\xF2&\x01\x00\x15#\x00\x00
PbyP Source Computer In Area Right	\xBE\xEF\x03\x06\x00\x86'\x01\x00\x12#\x00\x00
PbyP Source HDBaseT Area Left	\xBE\xEF\x03\x06\x00\xA2*\x01\x00\x15#\x11\x00
PbyP Source HDBaseT Area Right	\xBE\xEF\x03\x06\x00\xD6+\x01\x00\x12#\x11\x00
PbyP Source HDMI 1 Area Left	\xBE\xEF\x03\x06\x00\x02&\x01\x00\x15#\x03\x00

PbyP Source HDMI 1 Area Right	\xBE\xEF\x03\x06\x00v\x01\x00\x12#\x03\x00
PbyP Source HDMI 2 Area Left	\xBE\xEF\x03\x06\x00b"\x01\x00\x15#\x0D\x00
PbyP Source HDMI 2 Area Right	\xBE\xEF\x03\x06\x00\x16#\x01\x00\x12#\x0D\x00
PbyP Source Video Area Left	\xBE\xEF\x03\x06\x00b'\x01\x00\x15#\x01\x00
PbyP Source Video Area Right	\xBE\xEF\x03\x06\x00\x16&\x01\x00\x12#\x01\x00
PbyP Swap None	\xBE\xEF\x03\x06\x00\x01'\x06\x00\x16#\x00\x00
Picture Mode Board(Black)	\xBE\xEF\x03\x06\x00\xE3\xEF\x01\x00\xBA0 \x00
Picture Mode Board(Green)	\xBE\xEF\x03\x06\x00s\xEE\x01\x00\xBA0!\x00
Picture Mode Cinema	\xBE\xEF\x03\x06\x00\xB3\xF7\x01\x00\xBA0\x01\x00
Picture Mode Daytime	\xBE\xEF\x03\x06\x00\xE3\xC7\x01\x00\xBA0@\x00
Picture Mode Dicom Sim	\xBE\xEF\x03\x06\x00s\xC6\x01\x00\xBA0A\x00
Picture Mode Dynamic	\xBE\xEF\x03\x06\x00\xE3\xF4\x01\x00\xBA0\x04\x00
Picture Mode Natural	\xBE\xEF\x03\x06\x00#\xF6\x01\x00\xBA0\x00\x00
Picture Mode Standard	\xBE\xEF\x03\x06\x00\x83\xF5\x01\x00\xBA0\x06\x00
Picture Mode User-1	\xBE\xEF\x03\x06\x00\xE3\xFB\x01\x00\xBA0\x10\x00
Picture Mode User-2	\xBE\xEF\x03\x06\x00s\xFA\x01\x00\xBA0\x11\x00
Picture Mode User-3	\xBE\xEF\x03\x06\x00\x83\xFA\x01\x00\xBA0\x12\x00
Picture Mode Whiteboard	\xBE\xEF\x03\x06\x00\x83\xEE\x01\x00\xBA0"\x00
PIP Main Area Primary	\xBE\xEF\x03\x06\x002"\x01\x00\x05#\x00\x00
PIP Main Area Secondary	\xBE\xEF\x03\x06\x00\xA2#\x01\x00\x05#\x01\x00
PIP Position Bottom Left	\xBE\xEF\x03\x06\x00b"\x01\x00\x01#\x02\x00
PIP Position Bottom Right	\xBE\xEF\x03\x06\x00\xF2#\x01\x00\x01#\x03\x00
PIP Position Top Left	\xBE\xEF\x03\x06\x00\x02#\x01\x00\x01#\x00\x00
PIP Position Top Right	\xBE\xEF\x03\x06\x00\x92"\x01\x00\x01#\x01\x00
PIP Source Computer In Area Primary	\xBE\xEF\x03\x06\x00\xCE#\x01\x00\x04#\x00\x00
PIP Source Computer In Area Secondary	\xBE\xEF\x03\x06\x00F#\x01\x00\x02#\x00\x00
PIP Source HDBaseT Area Primary	\xBE\xEF\x03\x06\x00\x9E/\x01\x00\x04#\x11\x00
PIP Source HDBaseT Area Secondary	\xBE\xEF\x03\x06\x00\x16/\x01\x00\x02#\x11\x00
PIP Source HDMI 1 Area Primary	\xBE\xEF\x03\x06\x00>#\x01\x00\x04#\x03\x00
PIP Source HDMI 1 Area Secondary	\xBE\xEF\x03\x06\x00\xB6#\x01\x00\x02#\x03\x00
PIP Source HDMI 2 Area Primary	\xBE\xEF\x03\x06\x00^\x01\x00\x04#\x0D\x00
PIP Source HDMI 2 Area Secondary	\xBE\xEF\x03\x06\x00xD6"\x01\x00\x02#\x0D\x00
PIP Source Video Area Primary	\xBE\xEF\x03\x06\x00^\x01\x00\x04#\x01\x00
PIP Source Video Area Secondary	\xBE\xEF\x03\x06\x00xD6"\x01\x00\x02#\x01\x00
Power Off	\xBE\xEF\x03\x06\x00*\xD3\x01\x00\x00\x00\x00
Power On	\xBE\xEF\x03\x06\x00\xBA\xD2\x01\x00\x00\x01\x00
Video Mute Off	\xBE\xEF\x03\x06\x00\xFB\xD8\x01\x00 0\x00\x00
Video Mute On	\xBE\xEF\x03\x06\x00k\xD9\x01\x00 0\x01\x00
Volume Down Input Type Computer In	\xBE\xEF\x03\x06\x00z\xCD\x05\x00` \x00\x00
Volume Down Input Type HDBaseT	\xBE\xEF\x03\x06\x00v\xEB\x05\x00\xD5 \x00\x00
Volume Down Input Type HDMI 1	\xBE\xEF\x03\x06\x00>\xCD\x05\x00c \x00\x00
Volume Down Input Type HDMI 2	\xBE\xEF\x03\x06\x00xD6\xCF\x05\x00m \x00\x00
Volume Down Input Type LAN	\xBE\xEF\x03\x06\x00^\xCF\x05\x00k \x00\x00
Volume Down Input Type Video	\xBE\xEF\x03\x06\x00\x86\xCC\x05\x00a \x00\x00
Volume Up Input Type Computer In	\xBE\xEF\x03\x06\x00\xAB\xCC\x04\x00` \x00\x00
Volume Up Input Type HDBaseT	\xBE\xEF\x03\x06\x00\xA7\xEA\x04\x00\xD5 \x00\x00
Volume Up Input Type HDMI 1	\xBE\xEF\x03\x06\x00\xEF\xCC\x04\x00c \x00\x00
Volume Up Input Type HDMI 2	\xBE\xEF\x03\x06\x00\x07\xCE\x04\x00m \x00\x00

Volume Up Input Type LAN	\xBE\xEF\x03\x06\x00\x8F\xCE\x04\x00k \x00\x00
Volume Up Input Type Video	\xBE\xEF\x03\x06\x00W\xCD\x04\x00a \x00\x00

Appendix B. Query Commands

Aspect Ratio	\xBE\xEF\x03\x06\x00\xAD\xD0\x02\x00\x08 \x00\x00
Audio Mute	\xBE\xEF\x03\x06\x00u\xD3\x02\x00\x02 \x00\x00
AV Mute	\xBE\xEF\x03\x06\x00\xCD\xF0\x02\x00\xA0 \x00\x00
Closed Captions Channel	\xBE\xEF\x03\x06\x00qc\x02\x00\x027\x00\x00
Closed Captions Display	\xBE\xEF\x03\x06\x00\xC9b\x02\x00\x007\x00\x00
Closed Captions Mode	\xBE\xEF\x03\x06\x005c\x02\x00\x017\x00\x00
Device Status	\xBE\xEF\x03\x06\x00\xD9\xD8\x02\x00 ` \x00\x00
Eco Mode	\xBE\xEF\x03\x06\x00\x08#\x02\x00\x003\x00\x00
Filter Usage	\xBE\xEF\x03\x06\x00\xC2\xF0\x02\x00\xA0\x10\x00\x00
Freeze	\xBE\xEF\x03\x06\x00\xB0\xD2\x02\x00\x020\x00\x00
Input	\xBE\xEF\x03\x06\x00\xCD\xD2\x02\x00\x00 \x00\x00
Lamp Usage	\xBE\xEF\x03\x06\x00\xC2\xFF\x02\x00\x90\x10\x00\x00
PbyP Main Area	\xBE\xEF\x03\x06\x00I&\x02\x00\x13#\x00\x00
PbyP Main Size	\xBE\xEF\x03\x06\x00\xF1'\x02\x00\x11#\x00\x00
PbyP PIP Frame Lock	\xBE\xEF\x03\x06\x00y"\x02\x00\x17#\x00\x00
PbyP PIP Mode Select	\xBE\xEF\x03\x06\x00\x0D&\x02\x00\x10#\x00\x00
PbyP Source Area Left	\xBE\xEF\x03\x06\x00\xC1&\x02\x00\x15#\x00\x00
PbyP Source Area Right	\xBE\xEF\x03\x06\x00\xB5'\x02\x00\x12#\x00\x00
Picture Mode	\xBE\xEF\x03\x06\x00\x10\xF6\x02\x00\xBA0\x00\x00
PIP Main Area	\xBE\xEF\x03\x06\x00\x01"\x02\x00\x05#\x00\x00
PIP Position	\xBE\xEF\x03\x06\x001#\x02\x00\x01#\x00\x00
PIP Source Area Primary	\xBE\xEF\x03\x06\x00\xFD#\x02\x00\x04#\x00\x00
PIP Source Area Secondary	\xBE\xEF\x03\x06\x00u#\x02\x00\x02#\x00\x00
Power	\xBE\xEF\x03\x06\x00\x19\xD3\x02\x00\x00 \x00\x00
Video Mute	\xBE\xEF\x03\x06\x00\xC8\xD8\x02\x00 0\x00\x00